AWS WORLD SPORTS INNOVATION CUP 2026 · CHALLENGE 2

Diagonality

The Best of Both Worlds

Testing the tactical theory of diagonality — and turning it into a real-time, orientation-aware map — from 3D skeleton tracking.

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HOW IT WORKS

One pipeline, four orientation-aware stages

INPUT · 3D skeleton, 21 keypoints @ 50 Hz · measured head / shoulder / hip orientation of every player

1 Vision

Per-defender field of view.

Bekkers 120° cone, speed decay, torso occlusion.

2 Pitch Control

What each player can reach.

Anisotropic reach field; blob collapses in blind spot.

3 DOS Surface

Danger from a blind defender.

Control gained going diagonal vs straight.

4 Scanning Gate

Only what the player perceives.

2.5 s decayed memory — real-time, actionable.

VALIDATION ARM

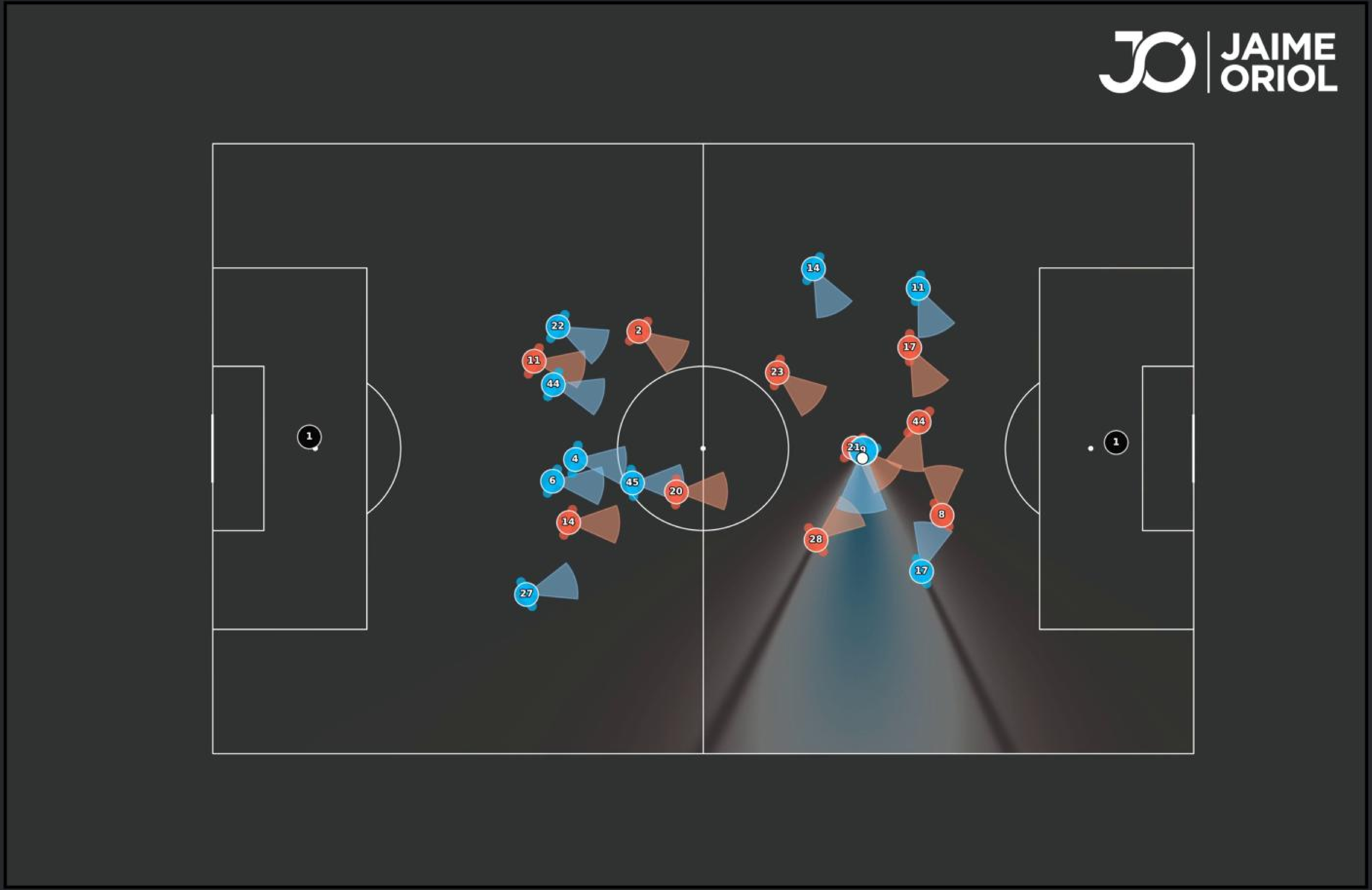
H1 perception · H2 bi-axial disruption · H3 receiver advantage · H4 progression–safety trade-off

6,923 actions
2,354 off-ball runs
theory holds

Skeleton geometry only — the DOS model never sees a goal, an xG, an xT or an outcome label.

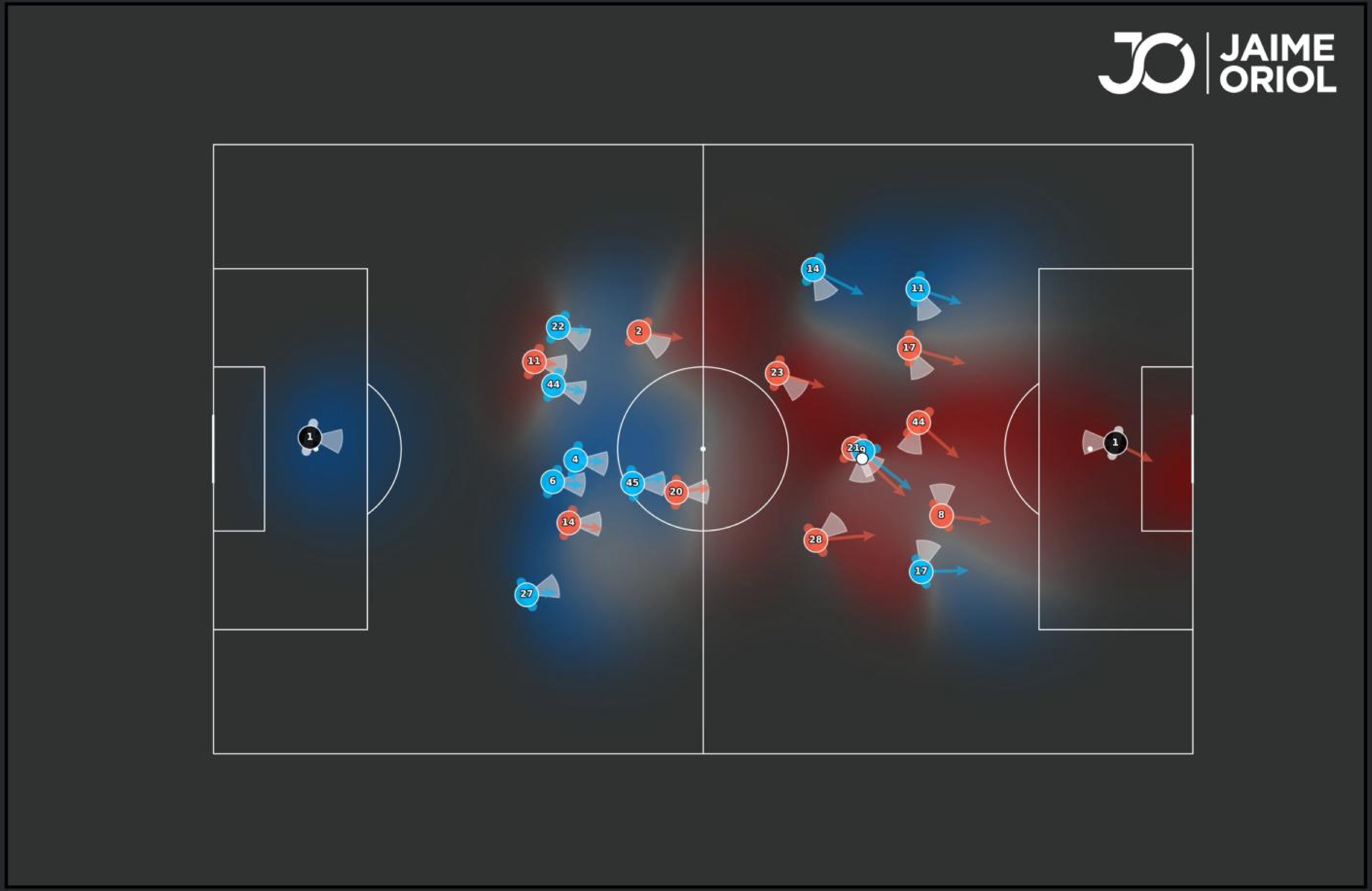
STAGE 1 — VISION

What every player can see



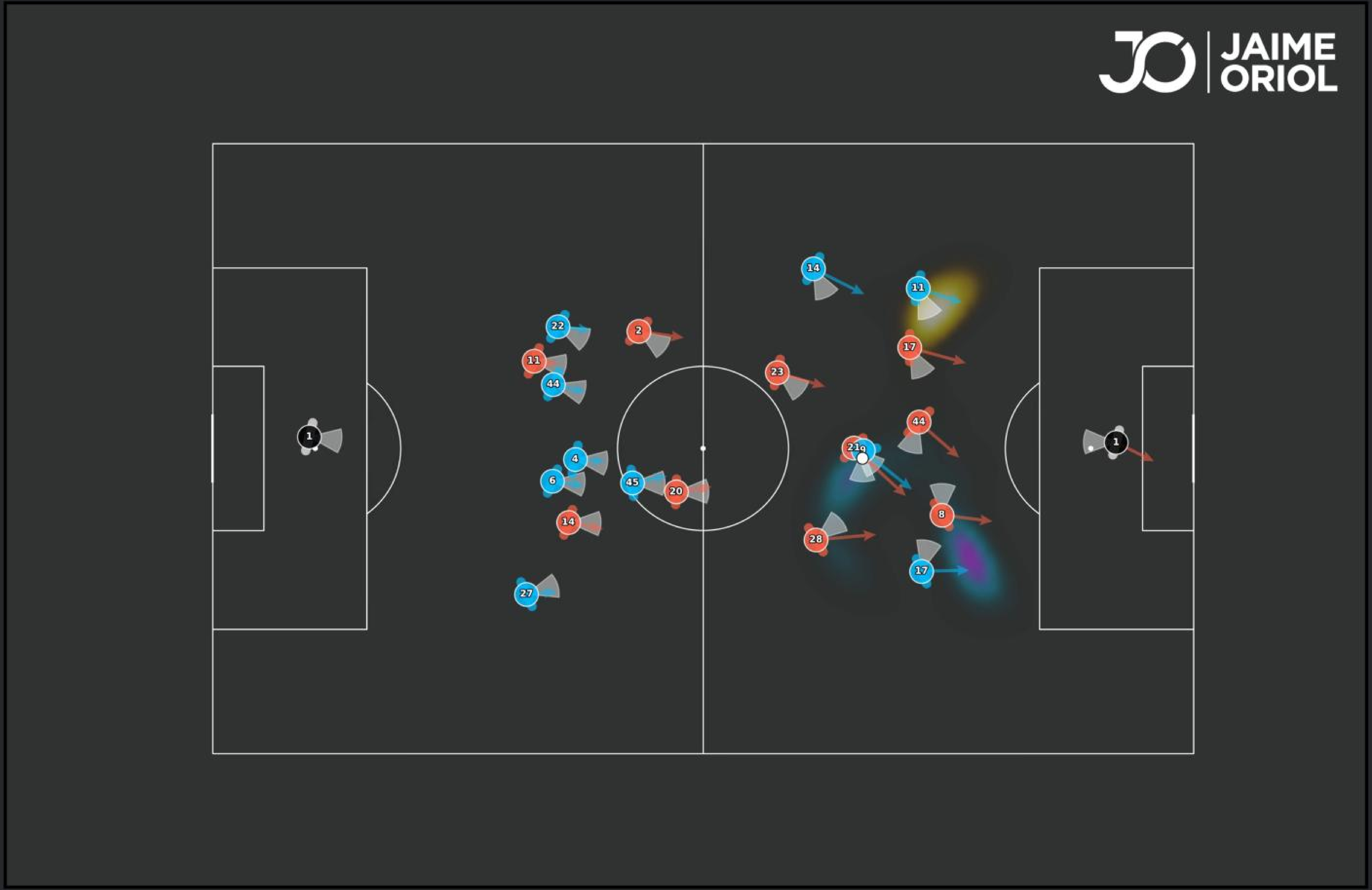
Probabilistic field of view: a 120° cone on the measured head yaw, speed-dependent decay, torso occlusion from real shoulder widths.

What every player can reach



Each player an anisotropic reach field — the blob collapses in a defender's blind spot, a literal hole in his control of space.

The Diagonal Opportunity Surface



Extra control bought going diagonal vs straight, gated to what the on-ball player perceives. Cyan: seen now. Amber: opportunity lost track of.